S M Masruk Uddin

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Residence permit: 9925068247 **Work permit:** French **Date of birth:** 2 Oct 1998 **Place of birth:** CHITTAGONG, Bangladesh **Nationality:** Bangladeshi

ABOUT ME

Master's graduate in Space Sciences from the Observatoire de Paris–PSL with experience in Galactic archaeology and looking for a PhD in Astrophysics. Worked on age determination of globular clusters using Bayesian inference and neural-network methods with HST, Gaia, and APOGEE data. Interested in galaxy formation and evolution through data-driven astrophysics.

EDUCATION AND TRAINING

[Sep 2023 – Jun 2025]

Master's in Space Science and Technology (International Research Track)

Observatoire de Paris- Paris Sciences & Lettres (PSL) University https://observatoiredeparis.psl.eu/international-research-track-4582.html

Address: 5, place Jules Janssen, F-92195 Meudon, 92195, Paris, France | **Field(s) of study:** Natural sciences, mathematics and statistics: ● *Physics* ● *Inter-disciplinary programmes and qualifications involving natural sciences, mathematics and statistics* | **Level in EQF:** EQF level 7

[Jan 2018 – Apr 2022]

Bachelor of Science in Physics (Hons)

Independent University, Bangladesh

City: Dhaka | **Country:** Bangladesh | **Field(s) of study:** Natural sciences, mathematics and statistics: ● *Physics ;* Minor in Computer Science

PUBLICATIONS

Revisiting the Age Scale of Galactic Globular Clusters. (In Prep.)

Authors: Uddin, M., Jash, R., Boin, T., Casamiquela, L., Reese, D. R., Haywood, M., | **Journal Name**: Target journal: A&A

RESEARCH

[Sep 2024 – Jul 2025]

Master 2- Lab Insertion Unit and Internship

Title: Revisiting the Age Scale of Galactic Globular Clusters

Supervisors: Misha Haywood, Laia Casamiquela, Daniel Reese

Lab: GEPI/LIRA

- Analyzed astrometric, spectroscopic, and photometric datasets (Gaia, APOGEE, and HST) to determine precise ages for 55 Galactic globular clusters (Entire HUGS catalog).
- Used Bayesian inference techniques with SPInS (Stellar Parameters INferred Systematically) to estimate stellar parameters using MCMC methods.
- Investigated the age–metallicity relation to distinguish in situ from ex situ globular clusters.
- Use of a Neural Network as an alternative and fast way to determine the age of globular cluster.

[Jan 2024 – Jun 2024]

Master 1- Internship

Title: Gravitational Waves from eccentric orbits around the central galactic black-hole Sgr A* .

Supervisor: Dr. E. Gourgoulhon (LUTH)

- Analyzed gravitational waveforms from compact objects in eccentric orbits around Sagittarius A* using the Black Hole Perturbation Toolkit. Compared waveforms for Schwarzschild and

Kerr black holes, examining the effects of spin, eccentricity, and semilatus rectum on signal strength.

- Conducted detailed analysis on detectability and found that LISA is capable of detecting such system, offering potential insights into compact object dynamics in strong gravitational fields, with some limits on the parameters.

[Sep 2021 – Dec 2021] **Senior-year Project in Physics**

Title: Infrared Divergence in Quantum Electrodynamics (QED) and other topics.

Supervisor: Dr. M Arshad Momen

- Explored Strong-Field (SF) physics with a focus on the Breit-Wheeler process and its divergent integrals, comparing it to charged scalar production.
- Reviewed fundamental concepts of relativistic quantum mechanics and strong field Quantum Electrodynamics (QED) and discussed emergence of infrared divergences in QED at the 1-loop level.

Other Projects

Title: The Effect of Second Order Spin in Periastronshift in the Kerr Spacetime. October 2020

DOI:10.13140/RG.2.2.35341.36326

- Calculated the periastron shift of a test body in a Kerr field. Analyzed contributions from both Newtonian and relativistic effects in a binary system. Investigated the possibility of measuring corrections induced by the square of the Kerr metric parameter a.
- Applied the results to the S2-SgRA* binary system and compared them with the Weyl class spacetime.

CONFERENCES AND SEMINARS

[1 Jul 2024 – 4 Jul 2024] **Semaine de l'Astrophysique Française** INP-ENSEEIHT, Toulouse, France

Organized by: French Society of Astronomy and Astrophysics (SF2A)

Poster: Revisiting the Age Scale of Galactic Globular Clusters

Authors: M. Uddin, R. Jash, T. Boin, L. Casamiquela, D. R. Reese, and M. Haywood

Abstract: https://journees.sf2a.eu/wp-content/uploads/inscription/Internship_Abstract_Masruk.pdf

ACADEMIC ACHIEVEMENTS

[Sep 2023 – Jun 2025] **International Mobility Grant by PSL Graduate Program in Astrophysics**

[8 Nov 2023] **Cum laude - 24th IUB Convocation**

[Sep 2018 – Dec 2021] **IUB Merit Scholarship**

Honors Lists

Dean's/Dean's Honour list (2018), Dean's/Dean's Merit list (2019), Vice Chancellor's list (2020), Dean's list(2021)

SKILLS

C++ | Python (computer programming) | Java (computer programming) | SLURM HPC | Mathematica Wolfram | TOPCAT (An interactive graphical viewer and editor for tabular data) | autoplot | LATEX, PyCharm, Visual Studio, MATLAB

LANGUAGE SKILLS

Mother tongue(s): Bengali

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

French

LISTENING A1 **READING** A1

SPOKEN PRODUCTION A1 **SPOKEN INTERACTION** A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SUMMER/WINTER SCHOOL

[25 Aug 2024 – 1 Oct 2024] **ICTP Physics Without Frontiers (Bangladesh)**

Teaching Assistant — School on Quantum Field Theory: https://indico.ictp.it/event/10720

[17 Dec 2019 – 21 Dec 2019] **India–Bangladesh Winter School in High-Energy Physics**

WORK EXPERIENCE

Department of Mathematical and Natural Sciences, BRAC University, Dhaka

City: Dhaka | **Country:** Bangladesh

[May 2022 – 31 Jan 2023] **Vice-Chancellor's Fellow & Course Contract Faculty**

Courses: PHY204 (Classical Mechanics & Special Relativity), MAT092 (Remedial Math), MAT110 (Differential Calculus & Coordinate Geometry), PHY111 (Mechanics, Properties of Matter), PHY112 (Electricity & Magnetism)

Physics Lab, Independent University, Bangladesh

City: Dhaka | **Country:** Bangladesh

[Sep 2018 – Mar 2022] **Student on Duty**

- maintenance of the Physics Lab and assisting teachers

VOLUNTEERING

[2021] **International Physics Olympiad (IPhO)— Invigilator** Bangladesh

[2017 – 2023] **Bangladesh Physics Olympiad — Trainer** Bangladesh